Image Mining – Practical Work

Part 2: Shot-cut detection

Zahra Soukhtedel

1. Introduction

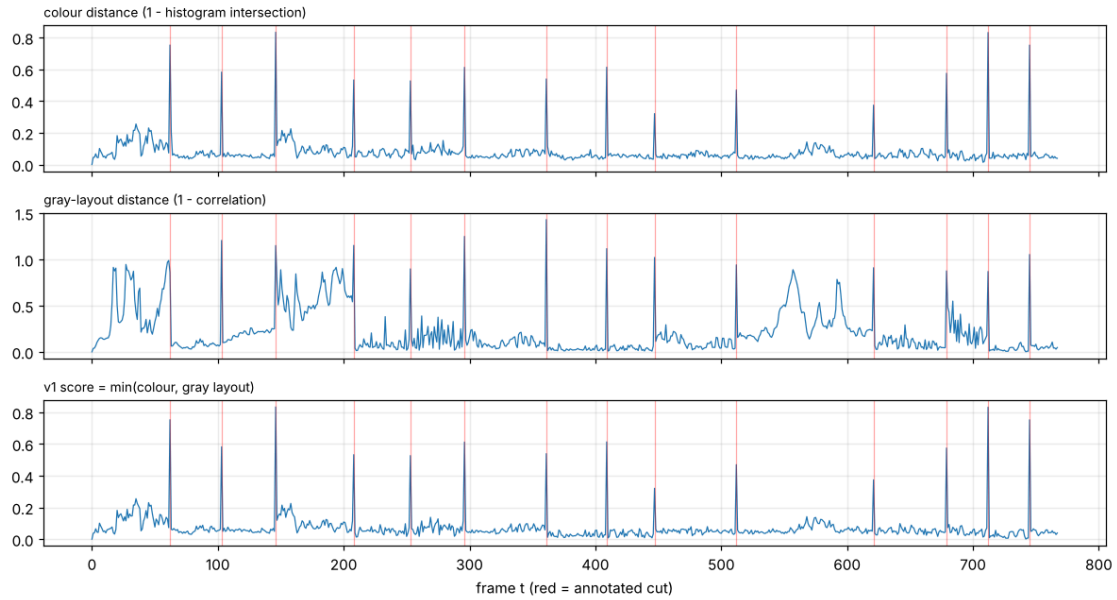
In this part I look for the hard cuts between shots in two video excerpts: The Matrix (768 frames, 14 annotated cuts) and Cosmos Laundromat (3 168 frames, 31 annotated cuts). I developed the method on Matrix and then ran it on Cosmos without changing anything, to see if it works on another video.

To compare the versions I used the same decision rule everywhere, so that only the score changes. Every frame is first shrunk to a 192×96 thumbnail (this is faster and hides noise). A boundary is a cut if its score is more than 4 times the median of the scores around it (±15 frames), if it is above 0.10, and if it is the largest value in 5 frames. I chose these values on Matrix and never touched them afterwards.

For the evaluation, a predicted cut is correct if it is at most 1 frame away from an annotated cut, and each annotated cut can be used only once. I count true positives (TP), false positives (FP) and false negatives (FN), and compute precision, recall and F1.

2. First approach (v1)

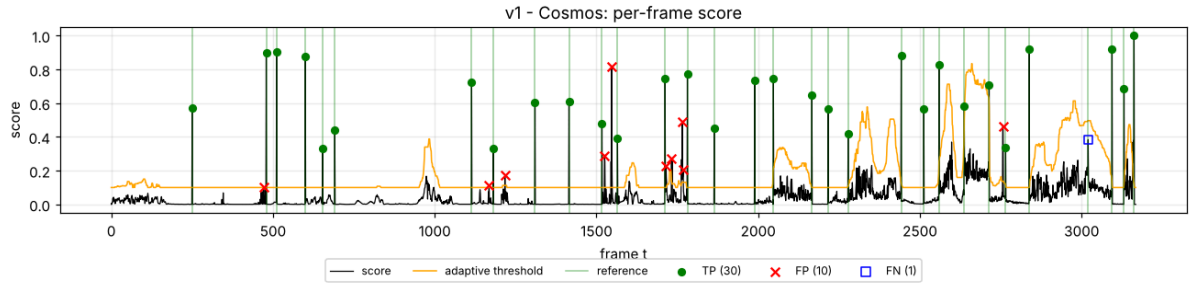
I describe each frame with a colour histogram (512 bins) and with a small grayscale image (32×16) that tells where things are. The histogram does not change much when the camera moves, and the grayscale image does not care about colours. For two frames the distance is the smaller of the two distances, and the score of a boundary compares frame t-1 and frame t. The idea is that a cut changes both colour and layout, camera motion changes only the layout, and a flash changes only the colours, so the minimum should only be high for real cuts.



On Matrix this works very well, as the figure shows. On Cosmos it does not work so well:

v1	TP	FP	FN	Precision	Recall	F1
Matrix	14	0	0	1.000	1.000	1.000
Cosmos	30	10	1	0.750	0.968	0.845

Cosmos, v1: score, threshold (orange), true positives (green), false positives (red crosses) and the missed cut (blue square)



v1 finds 30 of the 31 cuts but gives 10 false alarms, so the precision is only 0.75. Matrix did not show this at all, so I looked at the errors.

3. What goes wrong



Looking at the frames around the 10 false alarms, they are of two kinds. Three of them (f1525, f1547 and f2757) come from a single black frame in the middle of a shot: the picture is the same before and after, so it is not a cut, but a black frame changes both colour and layout, so the minimum cannot reject it when only two frames are compared. The other seven (f471, f1168, f1217, f1715, f1732, f1765, f1768) are lighting flashes: the colours change a lot but the objects stay in place. My grayscale layout is only a brightness image, so it changes as well.

So I needed a layout descriptor that does not care about the lighting, and a score that is not fooled by one strange frame.

4. Trial: HOG instead of the grayscale layout

A HOG (histogram of gradient orientations) tells where the edges are and in which direction. When the light changes, the edges do not move, so flashes should stop looking like cuts. I replaced the

grayscale layout by a HOG with 8 orientations and tried cells of 8, 12 and 16 pixels, and a version where every cell is normalised separately. Everything else stays as in v1.

Variant	Matrix TP/FP/FN	Matrix F1	Cosmos TP/FP/FN	Cosmos F1
v1: grayscale layout	14/0/0	1.000	30/10/1	0.845
2a: HOG, 8×8 px cells	14/0/0	1.000	29/6/2	0.879
2b: HOG, 12×12 px cells	14/0/0	1.000	28/6/3	0.862
2c: HOG 16×16, normalised per cell	13/0/1	0.963	24/4/7	0.814
2d: HOG, 16×16 px cells	14/0/0	1.000	29/5/2	0.892

All HOG versions remove false alarms (from 10 to between 4 and 6) but they also miss more cuts. This makes sense, because a descriptor that ignores lighting is also less sensitive to appearance in general. The per-cell normalisation is the worst (it even loses a cut on Matrix; I think it amplifies the noise in dark cells), and the small cells are a bit worse than 16 pixels. I kept the 16×16 cells: F1 0.892 against 0.845. So HOG alone already helps, but the black frames are still there.

5. Trial: compare several frames

A black frame lasts only one frame: the frame after it looks again like the frame before it. So I compare the two frames before the boundary with the two frames after it and I keep the smallest of the four distances. A single strange frame then gets a low score, while a real cut, where everything before is different from everything after, keeps a high one. I tried windows of 1+1, 2+2 and 3+3 frames with both layouts.

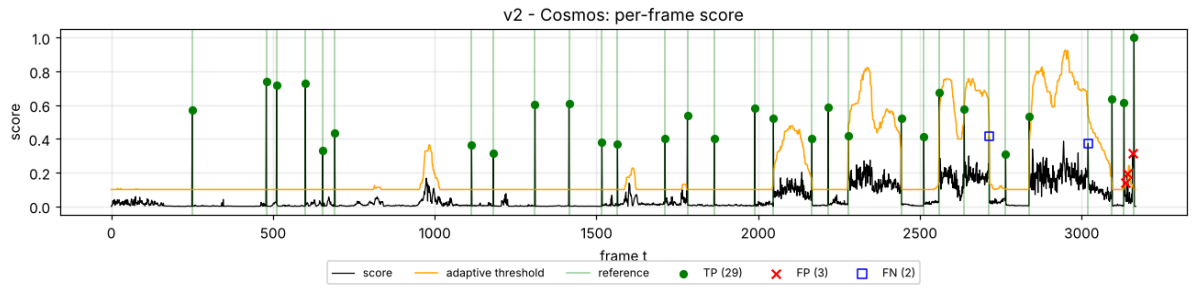
Variant	Matrix TP/FP/FN	Matrix F1	Cosmos TP/FP/FN	Cosmos F1
3a: gray, 1+1 (= v1)	14/0/0	1.000	30/10/1	0.845
3b: gray, 2+2	14/0/0	1.000	30/8/1	0.870
3c: gray, 3+3	14/0/0	1.000	30/7/1	0.882
3d: HOG, 1+1 (= 2d)	14/0/0	1.000	29/5/2	0.892
3e: HOG, 2+2	14/0/0	1.000	29/3/2	0.921
3f: HOG, 3+3	14/0/0	1.000	28/2/3	0.918

The window helps with both layouts: with gray the false alarms go from 10 to 8 (2+2) and 7 (3+3), with HOG from 5 to 3 and 2. With HOG, going from 2+2 to 3+3 removes one false alarm but also one real cut, so F1 does not improve (0.921 against 0.918). I kept 2+2, which has the best F1 and is the smallest window that handles a one-frame glitch.

6. Final version (v2)

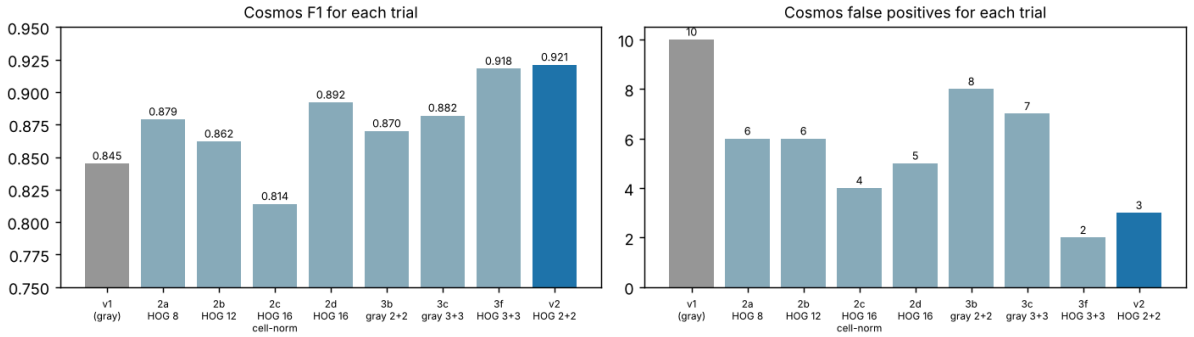
The final version puts the two improvements together, because each one fixes a different problem: HOG for the lighting changes and the 2+2 window for the black frames. The score of a boundary t is the smallest distance between a frame in {t-1, t-2} and a frame in {t, t+1}, where the distance between

two frames is the minimum of the colour distance and the HOG distance. The decision rule is exactly the one of v1.



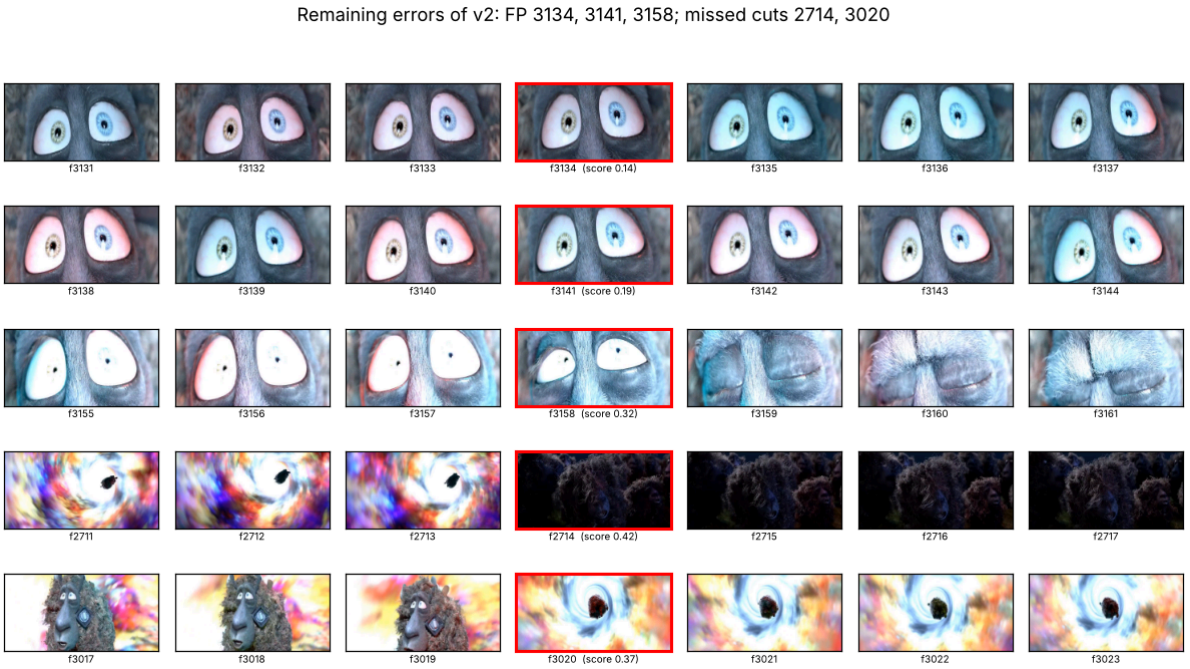
	TP	FP	FN	Precision	Recall	F1
v1 – Matrix	14	0	0	1.000	1.000	1.000
v1 – Cosmos	30	10	1	0.750	0.968	0.845
v2 – Matrix	14	0	0	1.000	1.000	1.000
v2 – Cosmos	29	3	2	0.906	0.935	0.921

Cosmos F1 and number of false positives for each trial (v1 on the left, v2 in dark blue on the right).



On Matrix both versions are perfect (the weakest cut scores 0.32 and the strongest non-cut 0.26 in both cases), so Matrix cannot tell them apart. On Cosmos, v2 is clearly better: the precision goes from 0.750 to 0.906 and F1 from 0.845 to 0.921. It removes all 10 false alarms of v1 (the black frames and the lighting changes) and in exchange it misses one more cut and has 3 new false alarms. The 29 cuts that are found are all at exactly the annotated frame. The two improvements add up (0.845 with v1, 0.892 with HOG only, 0.870 with the window only, 0.921 with both).

Remaining errors of v2. Rows 1–3: false positives f3134, f3141, f3158. Rows 4–5: missed cuts f2714 and f3020.



What is still wrong: the three false positives (scores 0.14, 0.19 and 0.32) are all in the last close-up of the eyes, where the colour of the eyes changes a lot and they move fast. The rest of that scene is very calm (local median around 0.03), so the threshold is only the floor of 0.10. The first two are borderline and the third one is a real fast movement. The two missed cuts (f2714 with score 0.42 and f3020 with 0.37) are in the busy, colourful end of the video, where the local median is about 0.13, so the threshold is around 0.51–0.53, above the score of the cut. f3020 was already missed by v1. f2714 was found by v1 (grayscale distance 0.71) but its HOG distance is only 0.45, because the two shots have similar edge directions, so here the HOG costs something.

About the threshold: for a ratio between 3 and 5, v2 keeps an F1 between 0.90 and 0.92 on Cosmos (v1: 0.81 to 0.85) and 1.00 on Matrix, so it also depends less on the exact value. With a ratio of 2 there are more false alarms, and with 8 or more weak cuts start to be missed. In general the adaptive threshold removes false alarms in busy scenes but costs recall in exactly those scenes.

7. Storyboard

For each shot found by v2 the storyboard (in the appendix) shows the frame before its start, its first frame, a representative frame, its last frame and the frame after its end, with the frame numbers. Matrix gives 15 shots and Cosmos 33. The first and the last frames belong to the neighbouring shots, so I can check each boundary by eye.

The representative frame is the one whose colour histogram is the closest (L1 distance) to the average histogram of the shot. I chose it because it is a real frame, it looks like the rest of the shot, and a fade or a flash at the border of the shot does not change it. A false positive shows up as two rows in a row with almost the same pictures, like the eye close-ups at the end of Cosmos.

8. Conclusion

v1 (colour histogram and grayscale layout on adjacent frames) is perfect on Matrix but only reaches F1 0.845 on Cosmos, because of 10 false alarms caused by black frames and lighting changes. After trying HOG layouts and different window sizes, the final version v2 (colour histogram, HOG layout, 2+2 window, same threshold rule) finds 14 of 14 cuts on Matrix and 29 of 31 on Cosmos with 3 false alarms (F1 0.921). It is better than v1 and less dependent on the threshold.

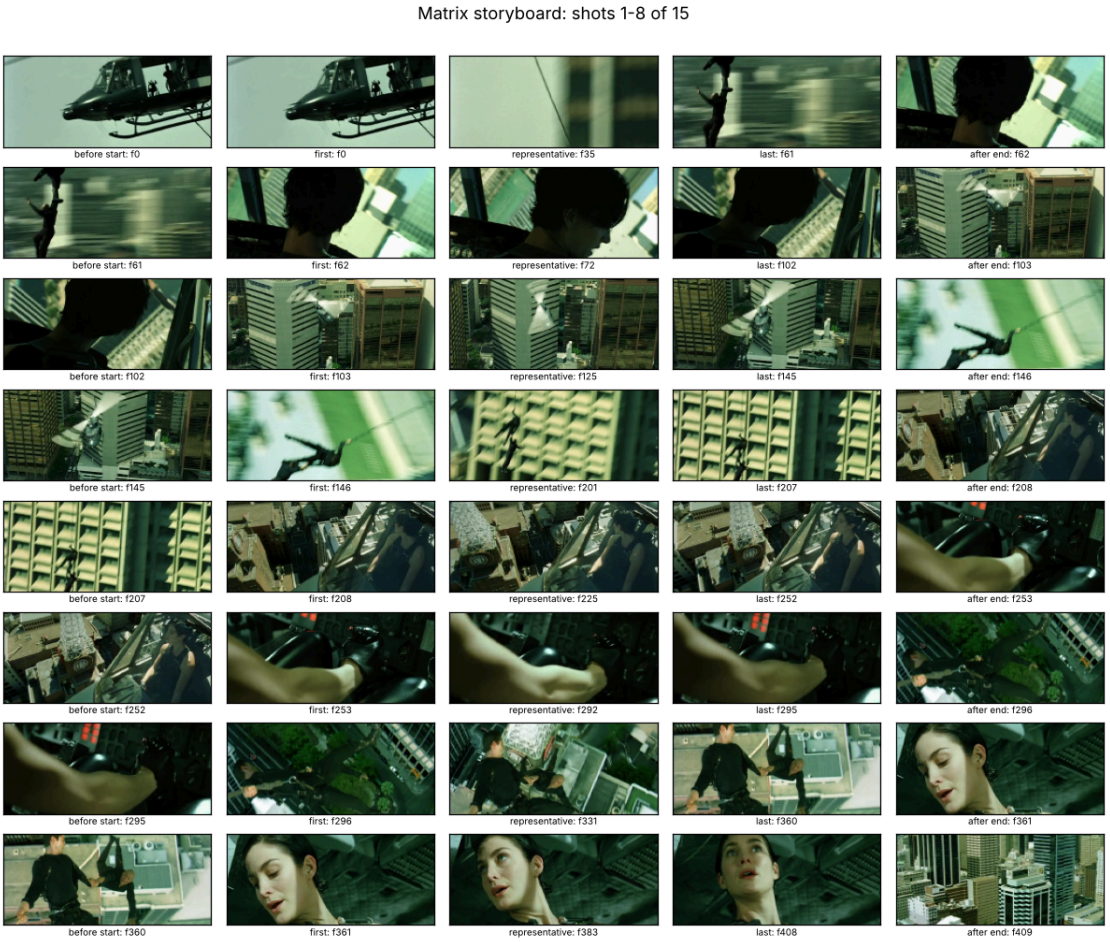
What remains difficult are strong colour changes inside a shot and weak cuts in very busy scenes. Things I did not try: a threshold based on something else than the local median, a detector for dissolves, and deep features.

One thing to be honest about: the parameters of the rule were chosen on Matrix and never changed, but I thought of the two changes of v2 after looking at the Cosmos errors of v1. So Cosmos is not a blind test for v2 any more. The real zero-shot result is the one of v1 (F1 0.845), and to measure v2 fairly I would need another annotated video.

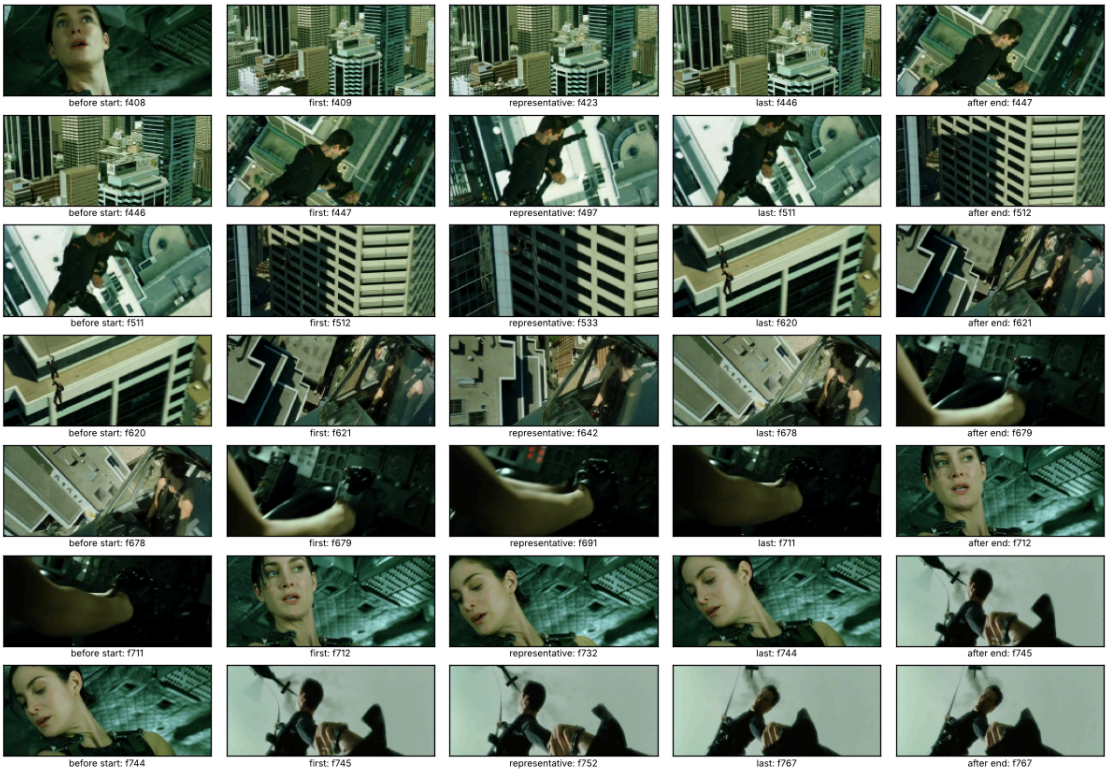
To run the code: Python 3.10 or newer, pip install av numpy scipy matplotlib pillow (OpenCV is used if PyAV is missing), put the two videos next to the notebook (or set VIDEO_DIR) and run all the cells.

Storyboards of v2

Each row: frame before the start, first frame, representative frame, last frame, frame after the end.



Matrix storyboard: shots 9-15 of 15



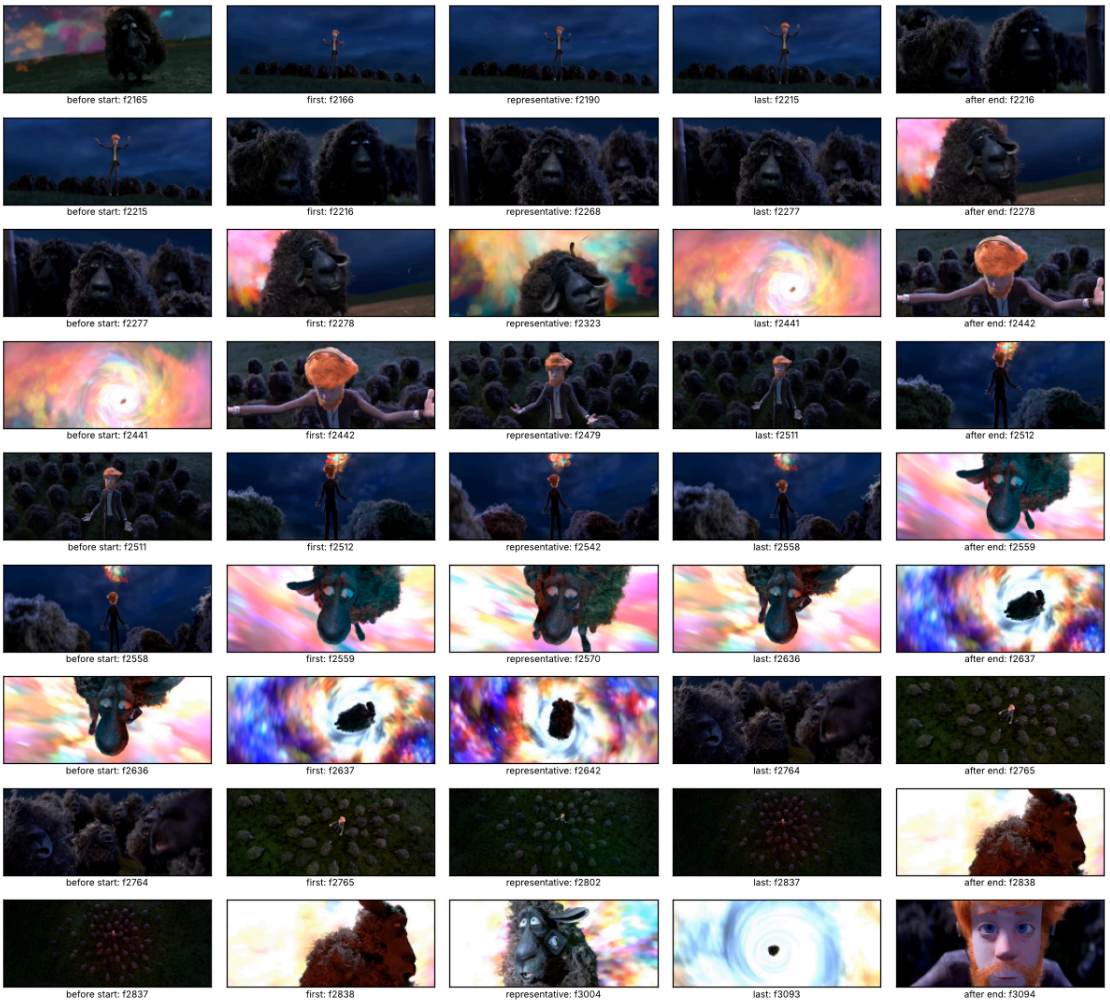
Cosmos storyboard: shots 1-9 of 33



Cosmos storyboard: shots 10-18 of 33



Cosmos storyboard: shots 19-27 of 33



Cosmos storyboard: shots 28-33 of 33

